

JIASEN LIU

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EDUCATION

University of Southern California (USC)

GPA: 3.97/4.00

M.A. in Mathematics

Jun 2024 - May 2026

B.S. in Mathematics & Minor in Computer Science

Aug 2022 - May 2026

Academic Interest: I am primarily interested in algebraic topology, and its connections to algebraic geometry, algebraic K -theory, and number theory. I also have a side interest in theoretical computer science and combinatorics, in particular computational social choice.

Relevant (graduate-level) Coursework (courses with * are audited):

- Fall 2023: Algebraic Topology, Enumerative Combinatorics
- Spring 2024: Algebra II*, Algorithms
- Fall 2024: Algebra I, Real Analysis, Toric Variety*, Advanced Algorithms, Voting Theory
- Spring 2025: Differential Geometry I, Algebraic Geometry, A^1 -Homotopy Theory, Functional Analysis
- Fall 2025: Lie Algebra, PDEs, Differential Geometry II, Representations for Finite Groups, Computational Complexity in Algebraic Combinatorics

I gained a Ph.D. Pass for the Topology Qualification Exam at USC.

RESEARCH EXPERIENCES

Undergraduate Researcher, Math REU, UChicago

Jun 2025 - Aug 2025

- Worked on Segal infinite loop space machine in equivariant setting and its connection to orthogonal G -spectra with fellow students K. Roke, H. Zhang, and K. Zhou, advised by prof. J.P. May.
- Studied homotopical algebra and simplicial presheaves, triangulated and derived categories, and algebraic K -theory, advised by Ajay Srivinishan.
- Wrote an expository paper on Milnor K theory and its connection to Galois cohomology.

Undergraduate Researcher, CS Theory Group, USC

Apr 2024 - Present

- Worked on projects in computational social choice and game theory with Yusuf Kalacyi, prof. David Kempe, and prof. Evi Micha.

Participant, Park City Undergraduate Summer School, PCMI

Jul 2024

- Attended 3-week lecture series focused on stable and motivic homotopy theory.
- Worked on classifying spaces and principal bundles with fellow students L. Borghi, S. Zhang, K. Zhou, and Z. Zobair.

Undergraduate Researcher, Polymath Jr. REU, Remote

Jun 2023 - Aug 2023

- Worked on a special kind of Diophantine equations with fellow students S. Arachichi, Q. Luan, and R. Marasinghe, advised by prof. Steven Miller and prof. Hùng Viet Chu.

ACADEMIC EXPERIENCES

Master Thesis, USC

Fall 2025 - Spring 2026

- Wrote an expository thesis on the Milnor conjecture and related topics, advised by prof. Aravind Asok.

Reading Course, USC

Spring 2025

- Studied Morse Theory and the h -Cobordism Theorem, advised by prof. Kyler Siegel.

Reading Course, USC

Spring 2025

- Studied Étale Cohomology, Characteristic Classes, and related topics, advised by prof. Aravind Asok.

- Directed Reading Program, USC** Fall 2023 & Fall 2024
 • Studied Commutative Algebra and Homological Algebra, advised by Ph.D. students at USC.
- Reading Course, USC** Spring 2024
 • Studied Cohomology Theories, Homotopy Theory, and an introduction to ∞ -categories, advised by prof. Joseph Helfer.

WORKSHOPS, SEMINARS & CONFERENCES

USC Topology Seminar	Nov 2024 - Present
USC CS Theory Lunch	Aug 2024 - Present
Fields Inst. Grad. School on Algebraic Topology	Jul 2025
International Conf. on Autonomous Agents & Multiagent Systems	May 2025
Preliminary Arizona Winter School: Local Fields	Sep 2024 - Nov 2024
PCMI Undergrad. Summer School: Motivic Homotopy Theory	Jul 2024
Connecticut Summer School in Number Theory	Jun 2024
USC Combinatorics Seminar: Asymptotic Algebraic Combinatorics	Jan 2024 - May 2024

PAPERS

1. **JL**, Kyle Roke, Hongyi Zhang, Keming Zhou, “*Symmetric Monoidal Categories and Pseudoalgebras over the Permutation Operad*”; in progress
2. **JL**, J. P. May, Kyle Roke, Hongyi Zhang, Keming Zhou, “*Multiplicative Equivariant Infinite Loop Space Theory Revisited*”; in progress
3. Yusuf H. Kalayci, **JL**, David Kempe, “*Full Proportional Justified Representation*”; Proceedings of *Intl. Conf. on Autonomous Agents and Multiagent Systems (AAMAS)*, 2025; arxiv: 2501.12015
4. Sujith U. Arachchi, Hùng Viet Chu, **JL**, Qitong Luan, Rukshan Marasingne, Steven J. Miller, “*On a pair of Diophantine Equations*”; arxiv:2309.04488

TEACHING EXPERIENCES

- Teaching Assistant, USC** Aug 2023 - Present
 • Fall 2023: MATH 425a (Mathematical Analysis I, Grader)
 • Spring 2024: CSCI 170 (Discrete Mathematics, TA)
 • Fall 2024: MATH 430 (Number Theory, TA), CSCI 270 (Intro to Algorithms, TA)
 • Spring 2025: CSCI 104 (Data Structures, TA)
- Counselor, Program in Math. for Young Scientists (PROMYS), Boston U.** Jun 2023 - Aug 2023
 • Graded the daily problem sets of 5 high school students in number theory and abstract algebra, gave presentations on relevant topics to around 100 audiences, and supervised a 5-student research group.

AWARDS & SCHOLARSHIPS

Archer Undergraduate Research Award, USC	Feb 2024
Academic Achievement Award, USC	Jan 2024
Dean’s List, USC	Aug 2022 - May 2025